DATA COMMUNICATION PROJECT

Digital Signal Encoder and Decoder

Project Specification | C++ Implementation

PROJECT CONTRIBUTORS

Aman Thappa
Enrollment: 2024BITE029

Millend Manhas
Enrollment: 2024BITE033

Project Objective

To develop a modular C++ application that demonstrates digital data generation, line coding, scrambling, and Pulse Code Modulation (PCM) encoding and decoding as part of a digital communication system.

Implemented Modules

1. Digital Data Generation G

enerates binary sequences, including random data and patterns containing four or eight consecutive zeros. The program also identifies the longest palindromic substring in the generated binary sequence.

2. Digital Line Coding I

mplements NRZ-L, NRZ-I, Manchester, Differential Manchester, and Alternate Mark Inversion (AMI) to represent binary data using signal levels or transitions.

3. Digital Scrambling I

mplements B8ZS (Bipolar with Eight-Zero Substitution) and HDB3 (High-Density Bipolar 3-Zero) for handling long runs of zeros in bipolar signal transmission.

4. PCM Encoder and Decoder A

ccepts analog samples in the range $[-10, 10]$, quantizes them into eight levels, and encodes each sample using three bits. The decoder reconstructs approximate sample values. The module also displays a staircase-style signal and exports reconstructed samples to `pcm_output.csv`.

5. Signal Visualization I

ncludes optional C++ visualization source files for NRZ signals and PCM reconstructed samples using the Raylib graphics library, which requires separate setup.

Technology and Assumptions

Language	C++ (C++11 or later)
Compiler	G++
Version Control	Git and GitHub
PCM Configuration	Eight quantization levels; three bits per sample; input range from −10 to 10
Output	Encoded digital signals, reconstructed PCM values, staircase display, and CSV output

Build and Execution

Run the following command from the repository root using a compatible G++ installation:

```
g++ -std=c++11 src/main.cpp src/data/data_generator.cpp
  src/pallindrome/pallindrome.cpp
  src/line_coding/line_coding.cpp src/pcm/pcm.cpp
  -o project.exe
```

On Windows PowerShell, start the application using:

```
.\project.exe
```

The application prompts the user to select digital or analog input. The selected workflow then processes the corresponding data. The build command should be checked against the final repository before use.

References and Acknowledgment

The primary reference is the project specification provided for the Data Communication course. OpenAI ChatGPT was used as an AI-assisted tool for programming guidance, debugging, concept explanations, and report preparation. The contributors are responsible for reviewing and understanding the code and for verifying the submitted implementation.